API GROUPS AND AUDITING

1. API COUNT SUMMARY (AUTHORITATIVE)

For a **robust but controlled v1 system**, MacroSight™ should expose:

Total APIs (logical groups): 11
Total endpoints (initial v1): ~55–65 endpoints

Breakdown:

Category	API Group
Authentication & Access	1
Users & Roles (Admin)	1
Sales Data	1
Products	1
External Indicators	1
Policy & Legislation	1
Forecast Models	1
Forecast Results	1
Scenario Simulation	1
Governance & Audit	1
System & Admin Ops	1

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2. AUTHENTICATION & ACCESS API

API Group: Auth API

Access: Public → Authenticated

Method	Endpoint	Purpose
POST	/api/v1/auth/login/	Authenticate user
POST	/api/v1/auth/refresh/	Refresh JWT
POST	/api/v1/auth/logout/	Invalidate session
GET	/api/v1/auth/me/	Get current user

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3. USERS & ROLES API (ADMIN-PRIVILEGED)

API Group: User Management API

Access: Admin only

Method	Endpoint	Purpose
GET	/api/v1/admin/users/	List users
POST	/api/v1/admin/users/	Create user
GET	/api/v1/admin/users/{id}/	User details
PUT	/api/v1/admin/users/{id}/	Update user
DELETE	/api/v1/admin/users/{id}/	Disable user
PUT	/api/v1/admin/users/{id}/role/	Assign role

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4. SALES DATA API

API Group: Sales API

Access: Analyst, Admin

Method	Endpoint	Purpose
POST	/api/v1/sales/upload/	Upload sales data
GET	/api/v1/sales/	Query sales
GET	/api/v1/sales/{id}/	Sale record
DELETE	/api/v1/sales/{id}/	Soft delete (admin)
GET	/api/v1/sales/export/	Export dataset

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5. PRODUCT API

API Group: Product API

Access: Analyst, Admin

Method	Endpoint	Purpose
GET	/api/v1/products/	List products
POST	/api/v1/products/	Create product

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Method	Endpoint	Purpose
PUT	/api/v1/products/{id}/	Update product
DELETE	/api/v1/products/{id}/	Remove product

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6. EXTERNAL INDICATORS API

API Group: Indicator API

Access: Analyst, Admin

Method	Endpoint	Purpose
GET	/api/v1/indicators/	List indicators
POST	/api/v1/indicators/	Manual indicator entry
GET	/api/v1/indicators/{id}/	Indicator details
PUT	/api/v1/indicators/{id}/	Update indicator
DELETE	/api/v1/indicators/{id}/	Remove indicator

Admin-Only Extensions

Method	Endpoint
POST	/api/v1/admin/indicators/sync/
PUT	/api/v1/admin/indicators/source/

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7. POLICY & LEGISLATION API (ADMIN-HEAVY)

API Group: Policy API

Access: Admin, Executive (read)

Method	Endpoint	Purpose
POST	/api/v1/policies/	Register policy
GET	/api/v1/policies/	List policies
PUT	/api/v1/policies/{id}/	Update policy
DELETE	/api/v1/policies/{id}/	Remove policy
POST	/api/v1/policies/{id}/impact/	Set impact score

API GROUPS

8. FORECAST MODEL API (ADMIN-PRIVILEGED)

API Group: Model Registry API

Access: Admin only

Method	Endpoint	Purpose
POST	/api/v1/models/register/	Register trained model
GET	/api/v1/models/	List models
GET	/api/v1/models/{id}/	Model metadata
PUT	/api/v1/models/{id}/	Update metadata
DELETE	/api/v1/models/{id}/	Retire model

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9. FORECAST RESULTS API

API Group: Forecast API

Access: Analyst, Executive

Method	Endpoint	Purpose
POST	/api/v1/forecasts/generate/	Generate forecast
GET	/api/v1/forecasts/	List forecasts
GET	/api/v1/forecasts/{id}/	Forecast details
GET	/api/v1/forecasts/{id}/explain/	Feature attribution

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10. SCENARIO SIMULATION API

API Group: Scenario API

Access: Analyst, Executive

Method	Endpoint	Purpose
POST	/api/v1/scenarios/run/	Run scenario
GET	/api/v1/scenarios/	List scenarios
GET	/api/v1/scenarios/{id}/compare/	Compare forecasts
DELETE	/api/v1/scenarios/{id}/	Remove scenario

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11. GOVERNANCE & AUDIT API (ADMIN-CRITICAL)

API Group: Governance API

Access: Admin, Auditor

Method	Endpoint	Purpose
GET	/api/v1/governance/model-runs/	List model runs
GET	/api/v1/governance/model-runs/{id}/	Run details
GET	/api/v1/governance/accuracy/	Accuracy metrics
GET	/api/v1/governance/drift/	Drift reports

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12. SYSTEM & ADMIN OPS API

API Group: System API

Access: Admin only

Method	Endpoint	Purpose
GET	/api/v1/system/health/	Health check
GET	/api/v1/system/config/	System config
PUT	/api/v1/system/config/	Update config
POST	/api/v1/system/retrain/	Trigger retraining
POST	/api/v1/system/backup/	Trigger backup

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13. ROLE → API ACCESS MATRIX (IMPORTANT)

Role	Access
Admin	All APIs
Analyst	Sales, Forecasts, Indicators, Scenarios
Executive	Read-only forecasts & scenarios
Auditor	Governance only

API GROUPS

1. DESIGN PRINCIPLES FOR AUDIT APIs

These APIs must be:

- **Read-only (GET only)**
- **Deterministic** (same input → same output)
- **Immutable views** over historical data
- **Heavily indexed & cached**
- **Role-restricted (Admin / Auditor)**
- **Human- and machine-readable**

They should **never** be used by the frontend dashboards.

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2. THE 5 AUDIT-GRADE APIs (AUTHORITATIVE SET)

API 1 — Forecast Lineage API

"Where did this forecast come from?"

Endpoint

GET /api/v1/audit/forecast-lineage/{forecast_id}/

Returns

- Forecast ID
- Model name & version
- Training period
- Input indicators used
- Policy events applied
- Timestamp of generation
- Hash of model artifact

Why this matters

- Full traceability
- Regulatory-grade provenance
- Zero ambiguity

API 2 — Model Run Registry API

"What models ran, when, and how well?"

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Endpoint

GET /api/v1/audit/model-runs/

Query Params

- model_name
- from_date
- to_date

Returns

- Model name
- Run timestamp
- Training window
- Accuracy metrics (MAPE, RMSE)
- Drift status
- Approved / deprecated flag

Why this matters

- Model governance
- Performance accountability
- Drift detection history

API 3 — Data Integrity Snapshot API

"What exact data did the model see?"

Endpoint

GET /api/v1/audit/data-snapshot/

Query Params

- snapshot_date
- dataset_type (sales | indicators | policies)

Returns

- Record counts
- Time coverage
- Missing data summary
- Outlier flags

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- Dataset checksum / hash

Why this matters

- Detect data tampering
- Verify completeness
- Support forensic analysis

API 4 — Indicator Impact Attribution API

"Which indicators actually influenced the forecast?"

Endpoint

GET /api/v1/audit/indicator-impact/{forecast_id}/

Returns

- Indicator name
- Contribution score (SHAP / attention weight)
- Direction (positive / negative)
- Magnitude ranking
- Confidence level

Why this matters

- Explainability
- Executive trust
- Compliance with AI governance standards

API 5 — Policy & Shock Effect API

"What was the quantified effect of policy or shocks?"

Endpoint

GET /api/v1/audit/policy-impact/{forecast_id}/

Returns

- Policy / legislation ID
- Effective date
- Severity score
- Forecast delta vs baseline

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- Duration of impact
- Affected products / regions

Why this matters

- Tax authority alignment
- Scenario validation
- Board-level decision support

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3. 5 APIS IS CORRECT COZ?

These 5 APIs collectively answer:

Question	API
Where did this forecast come from?	Forecast Lineage
How reliable is the model?	Model Runs
Was the data clean and complete?	Data Snapshot
Why did the forecast change?	Indicator Impact
What was the policy effect?	Policy Impact

Anything beyond this becomes **operational**, not **audit**.

API GROUPS